1 Summary

The GIVE IAM estimates the social cost of carbon (SCC) with four damage sectors (Rennert et al., 2022). We add a fifth damage sector for fire PM2.5 mortality. We parameterize key relationships to define a GIVE damage function identifying how fire PM2.5 mortality changes with global mean temperature (GMT) change.

Key parameters include 1) the relationship between GMT change and country-level average fire PM2.5 exposure per person and 2) relationship between PM2.5 exposure and mortality. We estimate the first while using an existing study for the second.

2 GMT change and fire PM2.5 exposure

For cell i in country c, year t in ten-year period p, and RCP scenario s, we have global grid cell level data for PM2.5 exposure, PM_{icts}, and fire PM2.5 exposure, fPM_{icts}. Decadal average data exists for a baseline period (2001-2010) and two future projections (2041-2050; 2091-2100) for different warming pathways (RCPs). For cell i, the fraction of PM2.5 attributable to fire is $\frac{fPM_{icts}}{PM_{icts}}$.

For country c, ten-year period p, and RCP scenario s, we define average annual *total* fire $PM_{2.5}$ exposure as:

$$\text{exposure}_{cps} = \frac{1}{10} \sum_{t \in p} \left(\sum_{i} fPM_{icts} \cdot p\bar{o}p_{ic} \right)$$

where denotes baseline population in grid cell i of country c. We define baseline population $p\bar{o}p_{ic}$ as the average population in grid cell i over the baseline period 2001-2010 [1]. To isolate the climate impact, population is held fixed across all periods and scenarios by using $p\bar{o}p_{ic}$ in place of time-varying population in all exposure calculations. We aggregate across grid cells for the total baseline population in country c, $p\bar{o}p_c = \sum_i p\bar{o}p_{ic}$.

We define per-capita fire $\text{PM}_{2.5}$ exposure as average exposure per person per year in country c over period p:

$$\text{exposure}_{cps}^{\text{per-capita}} = \frac{\frac{1}{10} \sum_{t \in p} \left(\sum_i fPM_{icts} \cdot p\bar{o}p_{ic} \right)}{p\bar{o}p_c}$$

This measure represents average exposure per person per year—population-weighted average—holding population fixed at baseline levels.

Separately for each country c, we estimate the following relationship across periods p and scenarios s:

$$\text{exposure}_{cps}^{\text{per-capita}} = \alpha_{c1} + \alpha_{c2} T_{ps}$$

[1] Note to self: Execute this population average $p\bar{o}p_{ic} = \frac{1}{10} \sum_{t \in \text{baseline}} pop_{ict}$ in R then multiply by the grid cell exposure level.

where T_{ps} denotes the change in global mean temperature (GMT) in period p under scenario s. The coefficient α_{c2} is interpreted as the change in per-capita fire $PM_{2.5}$ exposure (per person per year), holding population fixed at baseline levels, associated with a one-unit increase in GMT. We assume that this relationship is linear and constant over time.

3 PM2.5 exposure and mortality

The next key parameter relates PM2.5 exposure to moratlity. We source this parameter from Orellano et al. (2024)—a meta-analysis of fifty-three PM2.5 studies—and apply it to a standard health impact function, as defined in Anenberg et al. (2010).

To use Orellano's pooled relative risk (RR) in the Anenberg et al. (2010) health impact function, it must be converted from a per-10 μg/m^3 relative risk into a per-unit concentration–response coefficient β.

Anenberg et al. (2010) Equation 1 which defines the log-linear relationship between relative risk and concentration change:

$$\mathrm{RR} = e^{\beta \Delta X}$$

where β is the concentration–response factor and ΔX is the change[2] in concentration (μg/m^3). Rearranging for β:

$$\beta = \frac{\ln(\mathrm{RR})}{\Delta X}$$

Orellano et al. (2024) report a pooled RR of 1.095 (95% CI: 1.064–1.127) per 10 μg/m^3 increase in $PM_{2.5}$ for all-cause mortality. Substituting RR = 1.095 and $\Delta X = 10$ μg/m^3:

$$1.095 = e^{\beta \times 10}$$

$$\ln(1.095) = \beta \times 10$$

$$\beta = \frac{\ln(1.095)}{10} = \frac{0.09074}{10} \approx \frac{0.009075}{\mu\mathrm{g/m}^3} = 0.009075\ \mu\mathrm{g}^{-1}\mathrm{m}^3$$

The 95% confidence interval on β is implied from Orellano et al. (2024) as:

$$\beta_{\mathrm{lower}} = \frac{\ln(1.064)}{10} \approx 0.006203\ \mu\mathrm{g}^{-1}\mathrm{m}^3$$

[2]For studies applying a threshold above which PM2.5 exposure is considered fatal, ΔX is the difference between the exposure level and a given threshold. Since Orellano et al 2024 does not find any evidence for threshold effects, we consider ΔX to be the total exposure in a given cell.

$$\beta_{\text{upper}} = \frac{\ln(1.127)}{10} \approx 0.011956\ \mu\text{g}^{-1}\text{m}^3$$

This value of β is then substituted into Anenberg et al. (2010) health impact function (see below).

4 GIVE fire PM2.5 Damage Function

Anenberg et al. (2010) defines the change in mortality due to PM2.5 exposure as:

$$\Delta\text{Mort} = y_0 \left(1 - e^{-\beta \Delta X}\right) \text{Pop}$$

where y_0 is the baseline all-cause mortality rate, ΔX is the PM2.5 concentration exceeding a fatal threshold, and Pop is the exposed population. Ford et al. (2018) adopts the Anenberg et al. (2010) health impact function[3]. At grid cell level, Ford et al. (2018) computes:

$$\Delta\text{Mort All PM} = y_0 \left(1 - e^{-\beta \Delta X}\right) \text{Pop}$$

$$\Delta\text{Mort Fire PM} = (\Delta\text{Mort All PM}) \cdot (\text{fire PM } \mu\text{g}^{-1}\text{m}^3/\text{all PM } \mu\text{g}^{-1}\text{m}^3)$$

Thus, mortality attributable to fire PM2.5 equals the mortality from all PM2.5 times the fraction of total PM concentration due to fire.

We focus on fire PM2.5 exposure, rather than all PM2.5 exposure, assuming that these have the same mortality impact. However, studies suggest fire PM2.5 is more toxic (CITE THIS). For implementation in GIVE, we index by country c and year t,

$$\Delta\text{Mort}_{ct} = y_{ct} \left(1 - e^{-\beta \Delta X}\right) \text{Pop}_{ct}$$

Note, for forward-looking projections, the variables y_{ct} and Pop_{ct} are sourced from the GIVE model. In contrast to those variables noted above, these are time-varying future projections of population and mortality by country by year.

To associate the change in mortality with GMT change and to identify excess mortality rather than total mortality, we rewrite ΔMort_{ct} as:

$$(\text{Excess mortality})_{ct} = y_{ct} \left(1 - e^{-\beta \alpha_{c2} T_t}\right) \text{Pop}_{ct},$$

where $(\text{Excess mortality})_{ct}$ is the additional fire PM2.5 mortality for country c and year t

[3] In Ford et al. (2018), ΔX is the level of PM2.5 exceeding a threshold considered fatal. We will assume a threshold of zero, following Orellano et al. (2024). Thus, ΔX equals the PM2.5 expsoure in a given cell.

as a function on GMT change. Here, $\alpha_{c2}T_t$ is interpreted as the change in per-capita fire $PM_{2.5}$ exposure (per person per year) associated with GMT change in year t. GMT change is modeled within GIVE by the FAIR climate model.

Problem with above specification: Lower on the health impact function, seems to overestiamte fire PM mortality.

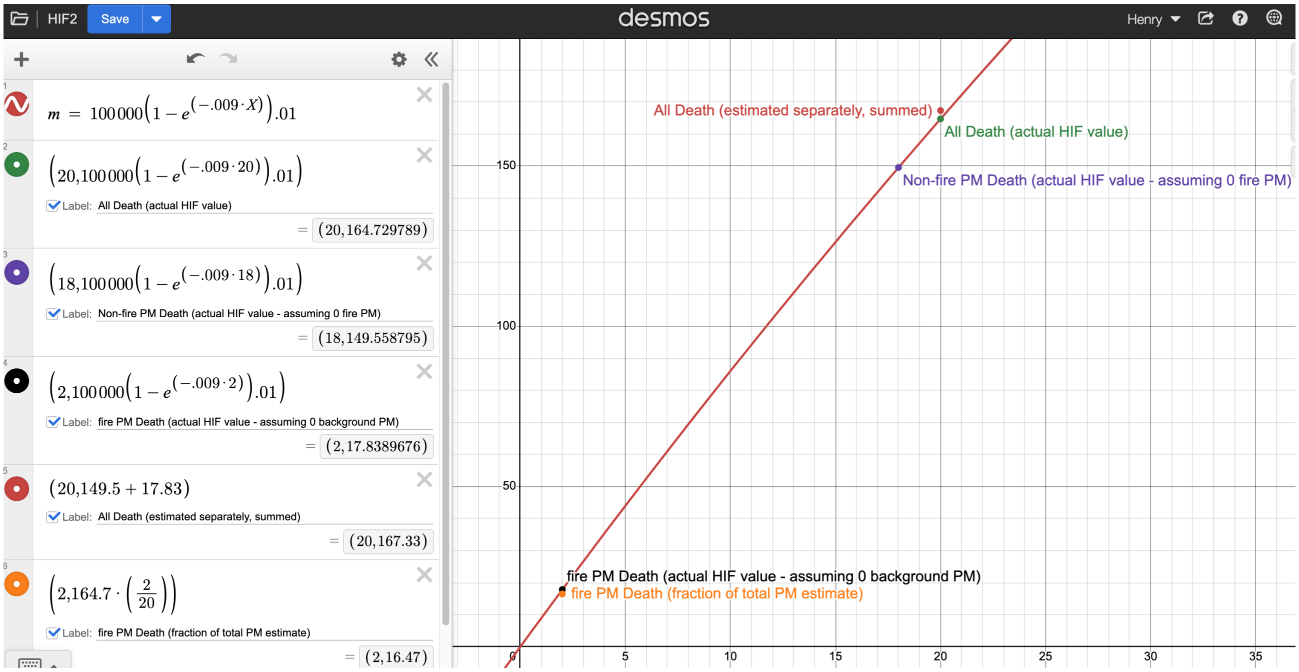

Figure 1:

instead, proposed ideal (CLAUDE) and using data might have (CLAUDE)

$$\Delta m_{ct} = \left[y_{ct}\left(1 - e^{-\beta\left(X^{\text{non-fire}}_{c,\text{pre-ind}} + \alpha_{c2} T_t\right)}\right)\text{Pop}_{ct}\right] - \left[y_{ct}\left(1 - e^{-\beta X^{\text{non-fire}}_{c,\text{pre-ind}}}\right)\text{Pop}_{ct}\right]$$

- Δm_{ct} — excess fire $PM_{2.5}$ mortality in country c, year t, attributable to all anthropogenic warming since pre-industrial
- y_{ct} — baseline all-cause mortality rate in country c, year t (from GIVE)
- β — concentration-response coefficient ($\mu g^{-1} m^3$); $\ln(1.095)/10 \approx 0.009075$
- $X^{\text{non-fire}}_{c,\text{pre-ind}}$ — population-weighted mean non-fire $PM_{2.5}$ in country c averaged over pre-industrial period (~1850–1900) ($\mu g/m^3$); computed as $X^{\text{total}}_{c,\text{pre-ind}} - X^{\text{fire}}_{c,\text{pre-ind}}$
- $X^{\text{total}}_{c,\text{pre-ind}}$ — population-weighted mean total $PM_{2.5}$ in country c averaged over pre-industrial period (~1850–1900) ($\mu g/m^3$)
- $X^{\text{fire}}_{c,\text{pre-ind}}$ — population-weighted mean fire $PM_{2.5}$ in country c averaged over pre-industrial period (~1850–1900) ($\mu g/m^3$)
- α_{c2} — regression coefficient: change in per-capita fire $PM_{2.5}$ ($\mu g/m^3$) per 1°C GMT change, country c; estimated from data spanning pre-industrial through 2091–2100 across RCP scenarios, anchored at $T \approx 0$
- T_t — FaIR GMT anomaly in year t relative to 1850–1900 (°C)
- Pop_{ct} — exposed population in country c, year t (from GIVE)

Figure 2:

$$\Delta m_{ct} = \left[y_{ct}\left(1 - e^{-\beta\left(X^{\text{non-fire}}_{c,1960} + \alpha_{c2} T_t\right)}\right)\text{Pop}_{ct}\right] - \left[y_{ct}\left(1 - e^{-\beta X^{\text{non-fire}}_{c,1960}}\right)\text{Pop}_{ct}\right]$$

- Δm_{ct} — excess fire $PM_{2.5}$ mortality in country c, year t, attributable to all anthropogenic warming since pre-industrial (approximated)
- y_{ct} — baseline all-cause mortality rate in country c, year t (from GIVE)
- β — concentration-response coefficient ($\mu g^{-1} m^3$); $\ln(1.095)/10 \approx 0.009075$
- $X^{\text{non-fire}}_{c,1960}$ — population-weighted mean non-fire $PM_{2.5}$ in country c averaged over 1960–1969 ($\mu g/m^3$); computed as $X^{\text{total}}_{c,1960} - X^{\text{fire}}_{c,1960}$
- $X^{\text{total}}_{c,1960}$ — population-weighted mean total $PM_{2.5}$ in country c averaged over 1960–1969 ($\mu g/m^3$)
- $X^{\text{fire}}_{c,1960}$ — population-weighted mean fire $PM_{2.5}$ in country c averaged over 1960–1969 ($\mu g/m^3$)
- α_{c2} — regression coefficient: change in per-capita fire $PM_{2.5}$ ($\mu g/m^3$) per 1°C GMT change, country c; estimated from data spanning 1960s through 2091–2100 across RCP scenarios
- T_t — FaIR GMT anomaly in year t relative to 1850–1900 (°C)
- Pop_{ct} — exposed population in country c, year t (from GIVE)

Figure 3:

5 OTHER NOTES

NOTES IN PROGRESS - NOT DEVELOPED

Whereas Mort_{ct} gives total mortality from total PM2.5 exposure ΔX, $(\text{Excess mortality})_{ct}$ gives excess mortality from the change in exposure per 1-degree GMT change.

not correct because not accounting for background PM2.5?

The correct quantity subtracts the background PM2.5? T_t and a counterfactual world with no warming ($T_t = 0$):

$$(\text{Excess mortality})_{ct} = y_{ct}\left(1 - e^{-\beta(X_c^{\text{baseline}} + \alpha_{c2}T_t)}\right)\text{Pop}_{ct} - y_{ct}\left(1 - e^{-\beta X_c^{\text{baseline}}}\right)\text{Pop}_{ct}$$

which simplifies to:

$$(\text{Excess mortality})_{ct} = y_{ct} \cdot e^{-\beta X_c^{\text{baseline}}}\left(1 - e^{-\beta\alpha_{c2}T_t}\right)\text{Pop}_{ct}$$

where X_c^{baseline} is country c's mean per-capita fire $\text{PM}_{2.5}$ exposure over the baseline period (2001–2010), defined as:

$$X_c^{\text{baseline}} = \frac{\frac{1}{10}\sum_{t\in\text{baseline}}\left(\sum_i fPM_{ict} \cdot p\bar{o}p_{ic}\right)}{p\bar{o}p_c}$$

This quantity is already defined in the exposure framework above as $\text{exposure}_{cps}^{\text{per-capita}}$ evaluated at $p = $ baseline, so no new data are required.

References

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